

## CONSTRUCTION (generator -- pre-gate filters)

Held substrate  
CRISPRi Perturb-seq  
CD4<sup>+</sup> T cells  
(aggregated tables)

Build universe A  
KD-gated +  
program-significant  
= 3,935 genes

Literature-novelty  
gate  
(rarity + curated  
association floor)

**22,039  
eligible  
(gene, disease)  
pairs**

## REFEREE (deterministic adjudication -- one verdict + receipt per pair)

Hop 0  
Knockdown QC

guide vs. NTC  
expression

Hop 1  
Effect

on-target DE  
(effect size, #DE)

Hop 2  
Program

Th1/Th2 shift  
(z, adj. p)

Hop 3  
Disease

module enrichment  
(OR, FDR)

supported

refuted

untested

each pair

**experimental receipts (referee-owned)**

**association receipt**  
(substrate-inherited stringency)

Cross-vendor independence supplied EXTERNALLY (S4.6): a 5-member replication lab  
(3 Opus-class + 2 Codex), 2 clean-room re-implementations -- unanimous pass.